

ICT/20/5819

THE FEDERAL UNIVERSITY OF TECHNOLOGY, AKURE
SCHOOL OF ENGINEERING AND ENGINEERING TECHNOLOGY
DEPARTMENT OF AGRICULTURAL AND ENVIRONMENTAL ENGINEERING

Name of Examination: 1ST SEMESTER, 2023/2024 SESSION

Course: AGE 301: ENGINEERING STATISTICS (2 UNITS)

Instruction: ATTEMPT ANY FOUR QUESTIONS

Time Allowed: 2 Hours 15 Minutes

Question 1

A study compared the tensile strength of fibers from three distinct banana cultivars for composite formation. Fifteen fibers were used, divided equally among the three cultivars, with five fibers per cultivar. Tensile strength was measured in newton per square meter (N/m^2), and the findings were recorded in a table.

Cultivar 1	Cultivar 2	Cultivar 3
23.3	36.0	25.2
22.9	38.0	25.9
24.1	37.2	23.5
23.6	38.3	24.8
21.7	37.8	24.2

When the significance level is set at 0.05, does the tensile strength mean differ among the three cultivars?

15 marks

Question 2

The following data represent the ages (in years) of people living in housing estate in Akure: ~~30, 31, 17, 16, 6, 2, 8, 43, 18, 18, 32, 33, 9, 18, 33, 19, 21, 13, 14, 13, 14, 6, 45, 52, 61, 23, 26, 14, 15, 14, 15, 27, 19, 36, 37, 11, 12, 11, 12, 20, 39, 40, 20, 63, 69, 64, 29, 28, 27, 15.~~
Present the above data in a frequency table using class interval of 10 and answer the following questions from the table.

- How many residents are aged between 11 and 30 years?
- How many residents are aged above 30 years?
- What is the probability that a person selected at random from the Estate will be less than 31 years old?

15 marks

Question 3 ✓

- a. The following data represent the weight of products manufactured in a factory (in kg).

Weight	45-54	55-64	65-74	75-84	85-94	95-104	105-114	115-124	125-134	135-144
No. of Products	1	3	5	18	33	25	21	12	5	2

Calculate the mean, median and mode.

8 marks

- b. The table below shows the frequency distribution of the scores of 106 students in AGE 301 test.

Score	0-10	10-20	20-30	30-40	40-50	50-60	60-70
No. of Students	1	3	5	18	33	25	21

Find the mean and mean deviation from the mean for the data.

7 marks

Question 4

- a. A manufacturer produces light-bulbs that are packed into boxes of 100. If quality control studies indicate that 0.5% of the light-bulbs produced are defective, what percentage of the boxes will contain; (a) no defective? $x=0$ (b) 2 or more defectives? $x \geq 2$ **9 marks**
- b. A random variable X is binomially distributed with mean 6 and variance 4.2. Find the number of the trials (n). **6 marks**

Question 5 ✓

In a dataset comprising ten products, each item had a diameter measured in meters represented by 'x' and its corresponding weight in kilograms denoted by 'y'

Product	1	2	3	4	5	6	7	8	9	10
Diameter (x)	170	169	177	178	178	175	182	185	163	160
Weights (y)	65	68	91	83	67	72	93	80	71	65

- a. Determine the Pearson correlation coefficient for this dataset
- b. Examine whether there is evidence to suggest a positive linear correlation between the heights and weights of the population, at a significance level of 5%
- c. Construct a linear model using the dataset and analyze its results with a significance level of 5%. **15 marks**